MIDTERM PROJECT: AI Agent Blueprint

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1. **Problem Statement**

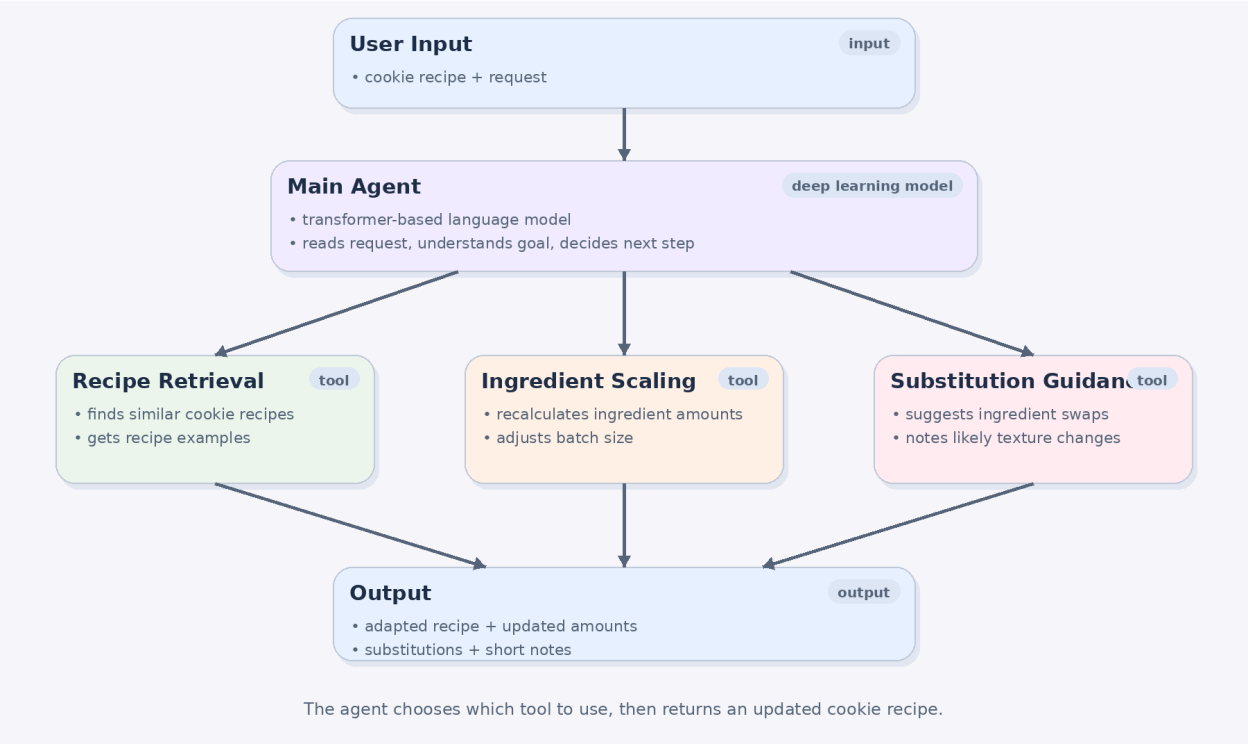
   As someone with a degree in pastry, I know from experience that adapting recipes can be frustrating and tricky. One problem I still encounter is having to manually recalculate ingredient amounts when changing batch sizes or making substitutions based on what I have available. Baking is a science where even small ingredient changes in a recipe can affect the texture, spread, and overall quality. My project focuses on designing an AI cookies recipe adaptation agent that helps users adjust common cookie recipes based on ingredient availability, batch size, and dietary needs. The intended users are beginners who want cleared guidance when making recipe changes. The problem matters because manual adjustments can be time-consuming and inconvenient. It can also be easy to get wrong which can lead to wasted time and ingredients. A helpful AI agent could make baking easier and more approachable.

2. **Chosen Option**

   I am choosing Option A: Single AI Agent for this project. I chose this option because I want to build one focused agent that solves a specific problem that I have personally dealt with. I think a single agent fits the idea better because the main goal is to have one agent that can understand the user's request, look at the recipe information, and give back an adjusted version with useful guidance. I also feel that Option A is the most realistic and manageable choice for me since I would consider myself still a beginner in deep learning. I hope to create something clear and practical instead of making the project too complicated.

3. Agent Architecture

My project uses a single AI agent that helps users adapt cookie recipes. The agent first reads the user's request such as changing the batch size, using substitute ingredients, or adjusting for dietary needs. Then it reasons through what changes need to be made. After that, it uses its tools such as recipe retrieval, ingredient scaling, and substitution lookup to make the adjustment. The final output is an updated cookie recipe with new ingredient amounts, suggested substitutions, and a short note about possible texture changes. The user can then ask follow up questions if they want another adjustment.



4. Deep Learning Connection

The main deep learning model for my project would be a transformer-based language mode. I chose this because my agent works with text (cookie recipes, ingredient lists, and

user requests). The agent needs to understand the recipe and then generate an updated version so a language model fits better than an image-based model.

This project connects to Module 04 and Module 05. Module 04 is relevant because it focused on NLP and sequence modeling. My project uses recipe text and user instructions as language input so that connection makes sense. Module 05 explains how transformers are used to understand context in text. My agent needs that ability so it can understand what the user wants changed and respond with a helpful recipe update. This project also relates to Module 09 since it is an AI agent that takes input, reasons about what to do, uses tools, and gives output.

5. **Agent Framework**

The agent framework I plan to use for the final build is LangChain. I chose LangChain because the framework guide mentions how LangChain is a strong choice for first time builders and single-agent projects. It also seems to have a large community with many tutorials that would be very helpful.

LangChain fits my project well because my cookie recipe adaptation agent needs one main agent that can read a user request, reason through recipe changes, and use tools like recipe retrieval, ingredient scaling, and substitution guidance.

6. **Tools and Data**

Recipe Retrieval Tool

- Retrieves relevant cookie recipe examples for the agent
- Supports the agent when comparing recipe structures and ingredient patterns

Ingredient Scaling Tool

- Recalculates ingredient amounts when the user wants to increase or decrease batch size
- Functions as a built-in calculation tool within the agent

Substitution Guidance Tool

- Suggests ingredient substitutions
- Helps explain how those substitutions may affect cookie texture and final results

USDA FoodData Central API (optional)

- Could be used for ingredient lookup and ingredient name standardization
- May help support more consistent substitution guidance
- *https://fdc.nal.usda.gov/download-datasets*

Food.com Recipes and Reviews dataset

- Main recipe dataset for the project
- Would be filtered to focus on common cookie recipes
- Review text may also help identify texture-related words such as *chewy*, *crispy*, or *soft*
- *https://www.kaggle.com/datasets/irkaal/foodcom-recipes-and-reviews/data*
- *https://www.kaggle.com/datasets/realalexanderwei/food-com-recipes-with-ingredients-and-tags*

Dessert Ingredients-to-Instructions dataset

- Optional secondary recipe dataset
- Could also be filtered to cookie recipes only
- *https://www.kaggle.com/datasets/luminolous/dessert-ingredients-to-instructions/data*

King Arthur Baking references

- Used as a small baking knowledge base
- Helpful for ingredient weights, baking measurements, and cookie texture guidance
- *https://www.kingarthurbaking.com/blog/2016/03/14/cookie-chemistry-2*

7. Build Plan

Week 1 (April 7-April 13)

- Finish the midterm blueprint and finalize the project scope so it stays focused on cookie recipe adaptation
- Select the agent framework and collect and organize main data sources

Week 2 (April 14 - April 20):

- Set up the basic single-agent system
- Implement the recipe retrieval tool and ingredient scaling tool
- Begin building the substitution knowledge base

Week 3 (April 21 - April 27)

- Build the substitution guidance tool
- Connect all tools to the agent workflow
- Test the agent on a few example cookie requests such as batch size changes and ingredient substitutions

Week 4 (April 28 - May 3)

- Refine the agent based on testing
- Complete the final cleanup and documentation
- Prepare the final demo and submit the project

8. Expected Challenges

One expected challenge is project scope. Baking and pastry are very broad and can be very complicated. Because of that, I decided to focus only on cookies so the project stays narrower and more realistic. This also matches the project advice to keep the idea focused instead of trying to do too much at once. Another challenge is that recipe adaptation is not always simple. Substitutions can change final results drastically. I also expect some difficulty in building the tools since I will need to make custom functions for recipe retrieval, ingredient scaling, and substitution guidance. Finally, I may need to simplify some parts of the project if the full version becomes too complex before the final deadline. I think this is a realistic plan and I have an advantage in that this is my domain of knowledge but I know I may need to adjust the scope as I build.